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Job Search in a Dynamic Economy

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1. INTRODUCTION

The last few years have witnessed an increasing interest in models of information and search with the seminal work of Stigler [24, 25] providing the basic theme. The major motivation for this work is twofold: to understand and guide the microeconomic behavior of individuals in a world of imperfect information, and to use this microeconomic theory as a basis for explaining macroeconomic phenomena like the Phillips curve. Many variants of the original Stigler model have been studied. These include sequential search policies [13, 15], the role of risk aversion [5, 9, 17, 28], adaptation to unknown distributions [20, 26] and a detailed analysis of barriers to search like discrimination and turnover [1, 2, 3, 21].[1] All of these variations retain the basic structure of the standard search model and testify to its enduring importance. It has contributed substantially to economic theory by introducing information costs and search into the highly unrealistic neoclassical models of perfect (costless) information. Nevertheless, the standard search model possesses severe limitations when viewed either as a normative guide to the individual job searcher or as a microeconomic rationale for macroeconomic behavior. The purpose of this paper is to criticize the standard model from both of these perspectives and construct alternative formulations that begin to repair some of the major difficiencies.

The limitations of the standard search methodology are best illustrated by quickly reviewing a stark but representative version. This version is presented in an infinite horizon setting with no discounting. A job offer X_i is received in the ith period of search, where X_i is a random variable with cumulative distribution function F. The distribution function F is known and invariant over time, and the X_i's are mutually independent. The job

[1] For a survey of this literature see [11].

searcher is assumed to retain the highest job offer, so that the return from stopping after the nth search is given by

$$Y_n = \max(X_1, \ldots, X_n) - nc,$$

where c is the cost per period of search. This cost includes all those out of pocket expenditures, like advertising and transportation, that are incurred each time a job offer is obtained.

The objective is to find a stopping rule that maximizes $E(Y_N)$ where N is the random stopping time, i.e., the random number of job offers before an acceptance.[2] Let R be the expected gain from searching according to the best stopping rule. Consider the first observation X_1. By definition of R, the first wage offer is accepted if it exceeds R and rejected (search continues) if it falls short of R. Clearly then for any wage offer x, the optimal policy has the form

$$\begin{array}{lll} \text{accept job} & \text{if} & x \geqslant R, \\ \text{continue search} & \text{if} & x < R. \end{array} \tag{i}$$

The expected return from this optimal policy is

$$\operatorname{Emax}(R, X_1) - c.$$

Since R was defined to be the expected return from pursuing the "best" stopping rule, this simple argument shows that the optimal expected return from the optimal stopping rule satisfies

$$R = \operatorname{Emax}(R, X_1) - c. \tag{ii}$$

which is equivalent to

$$c = \int_R^\infty (x - R)\, dF(x) \equiv H(R). \tag{iii}$$

This equation has a simple economic interpretation. The search cost c is, of course, the marginal cost of obtaining another job offer, whereas $H(R)$ is the expected marginal return from one more observation. The critical value R of a job offer is chosen to equate the marginal cost of one more offer with its expected marginal return.

The critical number R is called the *reservation wage*, and any search policy with a form given by (i) is said to possess the *reservation wage property*.

[2] The maximum expected gain is finite provided the expectation of the wage distribution exists. We assume this is the case so the existence of an optimal stopping rule is guaranteed. See [19].

The random variable N, the number of offers until R is exceeded, has a geometric distribution with parameter $p = 1 - F(R)$ and expected value $1/p$. Thus, the expected duration of search, "frictional unemployment," is an increasing function of the reservation wage. The job searcher will never commence search unless the expected return from unemployment is less than R, the expected return from search. This implies that labor force participation is an increasing function of the reservation wage.

The first limitation of this model is the assumption that the wage distribution is static, it is insensitive to the business cycle. Yet this model has been used extensively to "explain" dynamic behavior over the business cycle and to provide "optimal" strategies for job searchers who must contend with the vagaries of the business cycle. Clearly, the state of the economy is in constant flux and is an important determinant of search. In Section 2 we design a search policy that explicitly considers business cycle effects. A search policy that is optimal in a depressed economy is unlikely to be best in a boyuant economy. In particular, we expect the reservation wage to be lower in the former situation. It is precisely behavioral responses like this that generate macroeconomic phenomena. Indeed, most macroeconomic phenomena for which search models were designed manifest themselves only in a dynamic economy.

Our model envisages the economy changing according to a known Markov chain in which the probability of reaching a higher state in one transition is a nondecreasing function of the current state. Each state of the economy is characterized by a known distribution of wages. Assuming that the wage distribution is stochastically increasing in the state of the economy, our analysis demonstrates that for each state there is a reservation wage and this wage increases with the state of the economy. At this point it should be emphasized that ours is not an equilibrium model of search. Only the employee side of the search market is studied. Each searcher has his own reservation wage, and the heterogeneity of labor implies that there is no single wage below which firms cannot hire.

Section 3 presents the implications of the dynamic model for labor market behavior. These behavioral implications are contrasted with those of the static model.

Having introduced the business cycle it is obvious that the searcher may wish to vary his intensity of search as the economy changes. The optimal policy is set forth in Section 4. In contrast to the monotonicity of reservation wages with the state of the economy, it is somewhat surprising that when the individual can vary the rate of search, the optimal search intensity need not increase as the economy improves.

Another major criticism of the standard search model is its misuse in attempting to explain quits and layoffs. For example, the following argument has been developed to explain the Phillips curve (see [11]). Consider an increase in aggregate demand causing a shift in the demand for a firm's output. This in turn increases the demand for inputs, in general, and labor, in particular. Suppose this increase in demand is viewed as a temporary departure from some long-run trend. Wages rise and (this is the key assumption) searchers, *who are unaware of the increase in aggregate demand*, quickly find offers above R, the reservation wage in effect before the increase. Consequently unemployment declines as the rate of change in the price level increases. (The increase in aggregate demand induces a temporary increase in the rate of inflation.) A temporary decline in aggregate demand sets in motion corresponding forces on the downside. Wages decline and searchers, *being unaware of the decrease in aggregate demand*, do not change R as the wage offer distribution shifts to the left. Thus, unemployment increases as rate of change in inflation diminishes. This loose reasoning implies a negative short-run relationship between the rate of inflation and unemployment. Before questioning the logic of this "theory" consider the following empirical facts. Firms do not adjust wages in response to temporary changes in demand. Wages are sticky, especially on the downside, where employers adjust output by layoffs rather than wage reductions. The second empirical fact concerns the quit behavior during temporary recessions and temporary inflations. Quit rates decrease in recessions and increase in inflations. What does the standard search model "predict?" In recessions wages decline and workers quit because they mistakenly perceive steady wages elsewhere. Alternatively, in inflations workers should be less inclined to quit because they do not perceive that opportunities elsewhere are increasing at the same rate.

Having studied the "empirical implications" of the elementary search model, let us now investigate its theoretical foundations. The elementary search model was designed for a static economy, the wage distribution was assumed to be constant over time. Furthermore, employment constituted an absorbing state in this model. Once an employee accepts a wage offer he remains in that job at that wage until retirement. Similarly, employers were assumed to retain workers until their retirement. In spite of these limitations, the simple search model has been asked to explain a phenomenon that only makes sense in a dynamic economy. This model has also been admonished for not permitting layoffs and making incorrect predictions about quits. Yet the simple search model doesn't know what quits and layoffs are, and therefore, should not be used to make predictions about either. All of this suggests that search theory has been misused. The appropriate search model is one that specifies the search strategy as a function of the business cycle. The model discussed in Section 2 is a first step in this direction. If, in addition, an explanation of quit and layoff behavior is desired, then the dynamic search model must be modified to include both quits and layoffs.

In Sections 5, 6, and 7 we *begin* to develop a dynamic model that does include, respectively, quits, layoffs, and both quits and layoffs. The model is incomplete in that quits and layoffs are not viewed as jointly determined by both employer and employee. In addition, we assume that wages paid to employed workers do not change with the business cycle. The implicit contract theories of Gordon [8] *et al.* lend some credence to this assumption for certain classes of workers. Clearly, though, the assumption does not strictly hold. Typically, firms do alter wages in response to changing business conditions and if they do not, the precise arrangements underlying the implicit contract should be spelled out in detail. We have not yet resolved this problem, so rather than include an arbitrary adjustment mechanism in our already complex model, we assume the wage rate is constant. We also treat all workers alike and do not impose quit and layoff restrictions as a consequence of implicit employment contracts.

While our model is different from those previously employed in analyzing labor markets, it does include several of the previous models as special cases. We illustrate this in Section 3 for Salop's systematic search model and in Section 5 for Becker's specific human capital model.

2. A Simple Model of Search in a Dynamic Economy[3]

For expositional clarity, we begin with the simplest discrete time form of our model.

Let $\{1, 2, \ldots, K\}$, $K \leqslant +\infty$, be the set of states of the economy, and denote the distribution function of wages associated with state i by F_i. The discrete time discount factor is $\beta \leqslant 1$, and the search cost is $c > 0$. Naturally, we also assume that $\sup \mu_i < \infty$, where μ_i is the mean of F_i.[4] The economy changes according to a discrete time Markov chain, with one-step transition matrix $P = (P_{ij})$, which is independent of the sequence of offers drawn from $\{F_i\}$.[5]

The job offer available is the value of the random draw from last

[3] A similarly structured model has been used by Derman [7] in his study of optimal maintenance policies.

[4] The state space can be the entire real line provided $\lim_{x\to\infty} \mu_x < \infty$, where $\mu_x = \int_0^\infty y\,dF_x(y)$.

[5] Note this includes as a special case the economy moving according to a nonsymmetric random walk. In the continuous case, this would correspond to Brownian motion with a positive drift.

period's distribution function. If the searcher accepts an offer of x, the process terminates and the searcher is absorbed into the employment state for the n periods remaining in his working life. In this context, x can be regarded roughly as the discounted value of all future wages. If the searcher rejects x, he must pay a fixed price c for a draw from the distribution F_i. The economy then moves to a new state j according to P_{ij}. After this transition to state j, the searcher must decide whether to accept or reject the new offer y. The searcher is not allowed to retain rejected offers. The goal is to find a stopping rule which maximizes the β-discounted expected net benefits.

In order to ensure that the larger numbered states are preferable, we make two fundamental assumptions.

ASSUMPTION 1.[6] The distribution functions $\{F_i\}$ are stochastically increasing, that is

$$F_1(t) \geqslant F_2(t) \geqslant \cdots \geqslant F_K(t), \qquad \text{all } t \geqslant 0.$$

ASSUMPTION 2.[7] For any k,

$$\sum_{j=k}^{K} P_{ij} \text{ is nondecreasing in } i;^{8}$$

[6] Assumption 1 implies that $\mu_1 \leqslant \mu_2 \leqslant \cdots \leqslant \mu_K$ where μ_i is the mean of F_i, but, of course, not conversely. Moreover, it implies that $H_{i+1}(R) \equiv \int_R^\infty (x - R)dF_{i+1}(R) \geqslant \int_R^\infty (x - R)dF_i(R) \equiv H_i(R)$ for each R, so that if each state were absorbing and $\beta = 1$, the reservation wage associated with state $i + 1$ would be at least as large as that associated with state i.

[7] Assumption 2 rules out situations in which a high number state is inferior to a low number state. This situation could arise if the transition probability from a high state to a lower state was sufficiently large. Furthermore, define

$$\bar{\mu}_i^{(n)} = \sum_{j=1}^{K} P_{ij}^{(n)} \mu_j .$$

Then Assumptions 1 and 2 together imply not only that μ_i, the expected value of the immediately forthcoming offer, increases in i, but also that $\bar{\mu}_i^{(n)}$, the expected value of the $n + 1$st forthcoming offer, increases in i. In fact, the sequence $\langle X_{in} \rangle_{n=1}^{\infty}$ of offers received when starting from state i is stochastically increasing (term by term) in i. That is, $X_{i+1,n} > X_{i,n}$, all n. Thus, in this sense higher states are again preferable to lower states. The higher the state the more "buoyant" the economy.

[8] Assumptions 1 and 2 allow the following "bankruptcy" interpretation of the state of the system: Let i be the assets at the job searcher's disposal, and take X to be a nonnegative random variable (X need not be integer-valued nor even to have countable support since the state space may be uncountable) which represents the *total* return per dollar invested. Thus, if the value of the searcher's assets is i and he continues his search, he spends c on search and invests $i - c$, so that the value of his assets after

The importance of Assumptions 1 and 2 stems from the following facts: If f is a nondecreasing function, then

$$\int_0^\infty f(y)\, dF_i(y) \text{ is nondecreasing in } i \tag{1}$$

and

$$\sum_{j=1}^{K} P_{ij} f(j) \text{ is nondecreasing in } i. \tag{2}$$

We shall say we are in state (i, x) if the economy is in state i and the currently available job offer is x. Let $V_n(i, x)$ denote the maximal β-discounted expected return attainable when n periods remain and we are in state (i, x). Then $V_n(i, x)$ satisfies the recursive equation ($V_0 \equiv 0$)[9]

$$\begin{aligned} V_{n+1}(i, x) &= \max \left\{ x, -c + \beta \sum_{j=1}^{K} P_{ij} \int_0^\infty V_n(j, y)\, dF_i(y) \right\} \\ &\equiv \max\{x, R_n(i)\}. \end{aligned} \tag{3}$$

Clearly, $R_n(i)$ is the reservation wage $n + 1$ periods remain and the economy is in state i.

If we wish to interpret x as the wage rate rather than the discounted present value of all future wages, then x should be replaced by $x \sum_{j=0}^{n} \beta^j$ as the reward associated with stopping; thus, Eq. (3) becomes

$$V_{n+1}(i, x) = \max \left\{ x \sum_{j=0}^{n} \beta^j, R_n(i) \right\}. \tag{3a}$$

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search is given by $j = (i - c)X$. Now, if his assets fall below c he is not permitted any further search (for he is not permitted any reinvestment, this would be palatable if we interpret the search cost c as daily living expenses), and, hence, he will accept the last offer tendered. Assuming that the wage distribution sampled from does not depend upon his assets, except to account for the fact that he must permanently withdraw from the labor market if his assets should fall below c, the model obtains with (F_i) and (P_{ij}) given by $F_K = F_{K-1} = \cdots = F_c$; $F_{c-1} = F_{c-2} = \cdots = F_1 = F_0 =$ point pass at zero and

$$\begin{aligned} P_{ij} &= \text{Prob}\,[X = j/(i - c)], && \text{if } i > c, \\ &= 1 \quad, && \text{if } j = 0 \text{ and } i \leqslant c. \end{aligned}$$

Clearly, Assumptions 1 and 2 are satisfied. (See [5] for an alternative analysis of the effects of asset holdings on optimal search behavior.)

[9] Notice that for a single state economy this reduces, in the infinite horizon case, to the familiar relation

$$c = \int_R^\infty (x - R) dF(x), \qquad \text{with } \beta = 1.$$

Consequently, we can interpret $R_n(i)/\sum_{j=0}^{n} \beta^j$ as the reservation wage rate whereas $R_n(i)$ is merely the minimally acceptable discounted present value of all wages when $n + 1$ periods remain and the state of the economy is i.

THEOREM 1. *The optimal return $V_n(i, x)$ is nondecreasing in n, i, and x, so that the reservation wages $R_n(i)$ satisfy*

$$R_n(1) \leqslant R_n(2) \leqslant \cdots \leqslant R_n(K), \tag{4}$$

and

$$R_{n+1}(i) \geqslant R_n(i). \tag{5}$$

If $R_n(i)$ satisfies (3a), then

$$R_{n+1}(i)/\alpha_{n+1} \geqslant R_n(i)/\alpha_n, \tag{5a}$$

where $\alpha_n = \sum_{j=0}^{n} \beta^j$.

Proof. Note that $V_1(i, x) = x$, and assume $V_n(i, x)$ increases in both i and in x. Then

$$f(j+1) = \int_0^\infty V_n(j+1, y)\, dF_{i+1}(y) \geqslant \int_0^\infty V_n(j, y)\, dF_{i+1}(y),$$
$$\geqslant \int_0^\infty V_n(j, y)\, dF_i(y) = f(j),$$

by hypothesis and (1), respectively. Applying (1) and (2) to the above implies that $R_n(i+1) \geqslant R_n(i)$, so that $V_{n+1}(i, x)$ increases in both i and in x, establishing (4). Equation (5) is immediate from $V_n(i, x)$ nondecreasing in n.

Suppose (3a) holds. Clearly (5) holds. Now $V_2(i, x)/\alpha_1 \geqslant x\alpha_1/\alpha_1 = V_1(i, x)/\alpha_0$. Assume $V_n(i, x)/\alpha_{n-1} \geqslant V_{n-1}(i, x)/\alpha_{n-2}$ for all i and x, then since (5) holds and $-c/\alpha_n \geqslant -c/\alpha_{n-1}$, we obtain $R_n(i)/\alpha_n \geqslant R_{n-1}(i)/\alpha_{n-1}$ for all i. Therefore,

$$\begin{aligned} V_{n+1}(i, x) &= \alpha_n \max\{x, R_n(i)/\alpha_n\} \\ &\geqslant \alpha_n \max\{x, R_{n-1}(i)/\alpha_{n-1}\} \\ &= (\alpha_n/\alpha_{n-1}) \max\{x\alpha_{n-1}, R_{n-1}(i)\} = (\alpha_n/\alpha_{n-1})\, V_n(i, x). \end{aligned}$$

This completes the induction argument. Q.E.D.

Note too that $R_n(i)$ is a nondecreasing function of $\beta (0 \leqslant \beta \leqslant 1)$.

To treat the infinite horizon problem and obtain analogs of (3) and (4), it suffices to verify that $\langle V_n(K, x)\rangle_{n=1}^{\infty}$ is bounded for each x. To see this, merely note that $V_n(K, x) \leqslant V_n(x)$, where $V_0(x) = 0$

$$V_{n+1}(x) = \max \left\{x, -c + \beta \int_0^\infty V_n(y)\, dF_K(y)\right\},$$

and ($\mu_K \equiv \int_0^\infty [1 - F_K(t)]\, dt$)

$$0 \leqslant V_{n+1}(x) - V_n(x) \leqslant \beta \int_0^\infty [V_n(y) - V_{n-1}(y)]\, dF_K(y)$$

$$\leqslant \cdots \leqslant \beta^n \int_0^\infty [V_1(y) - V_0(y)]\, dF_K(y) = \beta^n \mu_K,$$

so that

$$V_n(x) = \sum_{j=1}^{n} [V_j(x) - V_{j-1}(x)] \leqslant \sum_{j=1}^{n} \beta^{j-1} \mu_K \leqslant \mu_K/(1 - \beta).$$

3. Labor Market Interpretation of the Simple Model

Since the reservation wage increases as the economy improves, there will be fewer dropouts in a bouyant economy. For suppose the economy is in state i and the tth individual has reservation wage $R^t(i)$. Suppose further that the return from not participating in the labor market exceeds $R^t(i)$. Then the optimal policy for the tth individual is to drop out. When the economy moves to a higher state j, this decision must be reconsidered, and since $R^t(j) \geqslant R^t(i)$, the tth individual may decide to search for employment. With nonwhites disproportionately represented in the pool of dropouts, we expect them to be disproportionately affected by an improving economy. The empirical literature on this subject is agreeable with this expectation (see [14, 23]).

It has been noted that the cost of search declines in a growing economy.[10] Notice that when the cost of search is state-dependent, with c_i denoting the cost of search when the economy is in state i, Theorem 1 remains true provided

$$c_1 \geqslant c_2 \geqslant \cdots \geqslant c_k.$$

This, of course, strengthens the negative relationship between economic growth and number of dropouts.

One would expect the length of search, the period of frictional unemployment, to decline as the economy improves. In general, this is not true. However, a simple but rather severe restriction on the Markov transition matrix, viz., $P_{ij} = P_j$, does produce the anticipated result.

More specifically, we wish to inquire as to the behavior of p_i, where p_i is the probability that a search from state i results in a job offer that will be

[10] For example, see Parsons [18] who argues that firms reduce search costs "by increasing advertising, arranging convenient interview times, paying moving expenses, etc."

accepted. To begin, define $q_{ij} = 1 - F_i(R(j))$, where $R(j)$ is the reservation wage (in the infinite horizon problem) for state j. Then q_{ij} is the conditional probability that a job offer resulting from a search from state i will be accepted given that the economy has moved to state j at the time the offer arrives. Hence, p_i satisfies

$$p_i = \sum_{j=1}^{K} P_{ij} q_{ij} .$$

If $P_{ij} = P_j$, then $p_i \leqslant p_{i+1}$. This follows since $q_{ij} \leqslant q_{i+1,j}$ as F_{i+1} is stochastically larger than F_i. This result states roughly that frictional unemployment decreases as the economy improves. More precisely, let Y_i be the number of offers tendered (until one is finally accepted) starting from state i, so that the random variable Y_i is precisely the period of frictional unemployment. Defining $p = \sum_j P_j p_j$ $(= \sum_j P_{ij} p_j)$, we have

$$P(Y_i = 1) = p_i , \; P(Y_i = k) = p(1-p)^{k-2}(1-p_i), \qquad k = 2, 3,...,$$

which, coupled with $p_i \leqslant p_{i+1}$, implies[11] that Y_{i+1} is stochastically smaller than Y_i. In other words, the period of frictional unemployment decreases stochastically in the state of the economy.

Recall that in the standard search model the job searcher is assumed to be unaware of the increase (decrease) in the offer distribution and consequently frictional unemployment declines (rises) as the economy improves. Here we make no such assumption about being fooled; the reservation wage increases (decreases) in a precise manner, and yet the behavior of frictional unemployment is, in general, ambiguous. If the standard model is to be consistent, the discouraged workers also will be unaware of changes in the wage distribution, and labor force participation will not vary as the economy changes.

Throughout this analysis we assume that the job searcher knows the wage distributions, the state of the economy, and the Markov chain. We do not address the important adaptive problems that arise when this information is imperfect. It is clear, however, that when the economy is in state j and the searcher believes it to be in i with $i < j$, the reservation wage will be set too low. This will lead to a reduction in frictional unemployment. The converse is true when $j < i$. Underestimation (overestimation) of the number of periods till retirement also reduces (increases) the reservation wage and, hence, frictional unemployment.

[11] Let $\alpha = \sum_{j=2}^{k} p(1-p)^{j-2} < 1$, so for $k > 1$,

$$\begin{aligned} P(Y_{i+1} \leqslant k) - P(Y_i \leqslant k) &= p_{i+1} + (1 - p_{i+1})\alpha - [p_i + (1 - p_i)\alpha] \\ &= (1 - \alpha)(p_{i+1} - p_i) \geqslant 0. \end{aligned}$$

In a recent paper, Salop [22, p. 191] states that "individuals are able to distinguish among firms *ex ante* and they sample *specific firms* in a systematic fashion rather than just sampling the job market in general." There, the reservation wage declines in the face of optimal systematic search. In addition, as the economy moves to inferior states (higher unemployment rates) the probability of a job offer declines and the reservation wage declines. The following version of our model yields similar results.

Assume that in addition to satisfying Assumption 2, $\{P_{ij}\}$ satisfies

$$P_{ii} = \lambda, \qquad \text{all } i. \tag{2*}$$

Equation (2*) implies that the economy spends the same expected time[12] in each state before changing to a new state.

Let F_{ij} be the distribution function for the jth offer received since the last re-entry into state i, $j = 1, 2, \ldots$. In place of Assumption 1, we assume

$$F_{i,j}(t) \geqslant F_{i+1,j}(t), \qquad \text{all } t, \text{ all } j. \tag{1*}$$

Thus, the jth offer received (since the last re-entry) in state i is stochastically smaller than the jth offer received in state $i + 1$, so that the sequence of offers tendered is most preferable (in a probabilistic sense) in the highest numbered states.

Furthermore, since the individual can distinguish among the various firms, we anticipate the result that his systematic search commences with the firm with the "best" offer distribution and proceeds at each step by selecting the firm with the best offer distribution from among those not yet searched. Labeling firms in ascending order according to this anticipated optimal systematic search, the offer distribution $F_{i,j}$ of the jth firm searched from state i is better than $F_{i,j+1}$. We will assume that $F_{i,j}$ is stochastically larger than $F_{i,j+1}$. For simplicity, further assume that

$$F_{ij} = (1 - q_i)\,\delta_0 + q_i F_j\,, \tag{1*a}$$

where q_i is the probability of receiving a (nonzero) offer resulting from a single search when in state i and δ_0 is the point mass at zero,

$$q_1 \leqslant q_2 \leqslant \cdots \leqslant q_K\,, \tag{1*b}$$

and

$$F_j(t) \leqslant F_{j+1}(t), \qquad \text{all } t, \text{ all } j. \tag{1*c}$$

Equations (1*a–c) comprise a specific simple parametrization of the stochastically ordered (increasing in the state i and decreasing in the firm

[12] In particular, (2*) implies that the number of job offers received before a change of state is a geometric random variable with parameter λ.

number j) nature within and between the sequences of job offers. (Of course, (1*a–c) imply (1*).) Then, it follows that

(a) the optimal search order is $\langle 1, 2,..., J\rangle$ for each i;

(b) the reservation wage $R_{i,j}$, from state i after j searches, is nondecreasing in i and nonincreasing in j;

(c) the expected duration of frictional unemployment decreases with i provided that $P_{ij} = P_j$ for $j \neq i$.

4. The Discrete Time Model with Variable Search Intensity

Typically the intensity of search is a variable that can be controlled by the job searcher. In general, it will depend on the searcher's preference for leisure, his wealth, and unemployment compensation. We do not incorporate these explicitly into our model, but instead place rather mild restrictions on the search cost function. The model was constructed as follows. Associated with search intensity μ, $0 \leqslant \mu \leqslant \bar{\mu}$, is (i) the search cost $c(\mu)$ and (ii) the probability of receiving a job offer $p(\mu)$, both of which are independent of the state of the economy; furthermore, given that an offer is received, it is independent of the search intensity. Naturally, both $c(\mu)$ and $p(\mu)$ are assumed to be nondecreasing. In addition, we assume that $c(0) = 0 = p(0)$ and $c(\cdot)$ is right continuous. For ease in presentation, we give $p(\cdot)$ the explicit representation $p(\mu) = \mu/(\mu + \lambda)$, where $\lambda > 0$. Since the searcher can have no impact on the economy, we assume that the Markov chain is independent of the search intensity chosen by the searcher.

It is now clear upon reflection that the analog of (3) is given by ($V_0 = 0$)

$$\begin{aligned} V_{n+1}(i, x) &= \max \left\{x, \max_{0 \leqslant \mu \leqslant \bar{\mu}} \left[-c(\mu) + (\beta/(\mu + \lambda)) \left(\lambda \sum_{j=1}^{K} P_{ij} V_n(j, 0) \right.\right.\right. \\ &\qquad \left.\left.\left. + \mu \sum_{j=1}^{K} P_{ij} \int_0^\infty V_n(j, y)\, dF_i(y)\right)\right]\right\} \\ &= \max\{x, R_n(i)\}. \end{aligned} \tag{6}$$

We are interested in the behavior of not only the reservation wage $R_n(i)$ but also $\mu_n(i)$, where $\mu_n(i)$ maximizes $R_n(i)$, that is, where $\mu_n(i)$ is the optimal search intensity when $n + 1$ periods remain and the state of the economy is i.

Theorem 2. *With variable search intensity, $V_n(i, x)$, as given in* (6), *is nondecreasing in n, i, and x, and the reservation wages satisfy* (4) *and* (5).

Proof. To begin, observe that

$$
\begin{aligned}
&[R_n(i+1) - R_n(i)](\mu+\lambda)/\beta \\
&\geqslant [-c(\mu_n(i)) + c(\mu_n(i))] + \lambda \sum_{j=1}^{K} (P_{i+1,j} - P_{i,j})\, V_n(j, 0) \\
&\quad + \mu_n(i) \left[\sum_{j=1}^{K} P_{i+1,j} \int_0^\infty V_n(j, y)\, dF_{i+1}(y) - \sum_{j=1}^{K} P_{i,j} \int_0^\infty V_n(j, y)\, dF_i(y) \right] \\
&\geqslant 0.
\end{aligned}
$$

The second term on the R.H.S. is greater than or equal to zero by (2) and the induction hypothesis that $V_n(i, x)$ is nondecreasing in both i and x while the third term exceeds zero by (1), (2) and the induction hypothesis. The first inequality follows from $\mu_n(i)$ being suboptimal from state $i+1$. Q.E.D.

In light of the fact that the reservation wage $R_n(i)$ increases with the state of the economy, it would come as no surprise if the optimal search intensity $\mu_n(i)$ increased with i. In general, this result is not true. In fact, all possible orderings of $\{\mu_n(i)\}$, with the proviso that $\mu_n(K) \geqslant \mu_n(1)$, are possible. Moreover, it is not even necessary that $\mu_n(i)$ be monotone in n. (As will be clear, the added assumption of convexity of $c(\cdot)$ is not relevant in eliminating this seeming pathology.)

EXAMPLE 1. If $P_{ij} = P_j$, for all i, then

$$\mu_n(1) \leqslant \mu_n(2) \leqslant \cdots \leqslant \mu_n(K), \tag{7}$$

and

$$\mu_1(i) \geqslant \mu_2(i) \geqslant \cdots \geqslant \mu_n(i) \geqslant \mu_{n+1}(i) \geqslant \cdots \geqslant \mu(i), \tag{8}$$

where $\mu(i)$ is the optimal search intensity for the infinite horizon problem.

Proof. We can reformulate (6) as follows ($V_0 \equiv 0$ and $V_n(i)$ is to be interpreted as the maximal β-discounted expected return attainable when n periods remain, the state of the economy is i, and no offer is currently available):

$$
\begin{aligned}
V_{n+1}(i) = \max_{0 \leqslant \mu \leqslant \bar{\mu}} \Big\{ & -c(\mu) + (\lambda/(\mu+\lambda))\beta \sum_{j=1}^{K} P_{ij} V_n(j) \\
& + (\mu/(\mu+\lambda)) \int_0^\infty \max\left[\beta y, \beta \sum P_{ij} V_n(j)\right] dF_i(y) \Big\}.
\end{aligned} \tag{9}
$$

In this case, if $P_{ij} = P_j$ for all i, we have, setting $V_n = \beta \sum_{j=1}^{K} P_{ij} V_n(j)$,

$$V_{n+1}(i) = \max_{0 \leqslant \mu \leqslant \bar{\mu}} \Big\{ -c(\mu) + (\lambda/(\mu + \lambda)) V_n + (\mu/(\mu + \lambda)) \times \Big[V_n + \int_{V_n/\beta}^{\infty} (\beta y - V_n) \, dF_i(y) \Big] \Big\}$$

$$= \max_{0 \leqslant \mu \leqslant \bar{\mu}} \Big\{ -c(\mu) + (\mu/(\mu + \lambda)) \int_{V_n/\beta}^{\infty} (\beta y - V_n) \, dF_i(y) \Big\} + V_n .$$

It now follows from Eq. (1), with $f(y) = 0$ for $y \leqslant V_n/\beta$ and $f(y) = \beta y - V_n$ for $y > V_n/\beta$, that $\mu_n(i + 1) \geqslant \mu_n(i)$, for $i = 1, 2, \ldots, K - 1$ and all $n \geqslant 1$.

Finally, note that since $\int_x^{\infty} (y - x) \, dF(y)$ decreases as x increases, V_n nondecreasing in n would imply that

$$c_n \equiv \int_{V_n/\beta}^{\infty} (\beta y - V_n) \, dF_i(y) \geqslant \int_{V_{n+1}/\beta}^{\infty} (\beta y - V_{n+1}) \, dF_i(y) \equiv c_{n+1} .$$

Since $c(0) = 0$, V_n is, in fact, nondecreasing in n. Consequently, we can write

$$V_{n+1}(i) = \max_{0 \leqslant \mu \leqslant \bar{\mu}} \{ -c(\mu) + (\mu/(\mu + \lambda)) \, c_{n+1} + (\mu/(\mu + \lambda))(c_n - c_{n+1}) \} + V_n .$$

Thus, the maximand of $V_{n+1}(i)$ is the sum of the maximand of $V_{n+2}(i)$ and and an increasing function of μ. Hence, we can conclude that (8) holds.

Q.E.D.

EXAMPLE 2. $\mu(K) \geqslant \mu(1)$.

Proof. Let $V(i) = \lim_n V_n(i)$, where $V_n(i)$ satisfies (9), so that $V(i)$ is the optimal return for an infinite horizon when the economy is in state i. Then

$$V(i) = \max_{0 \leqslant \mu \leqslant \bar{\mu}} \Big\{ -c(\mu) + (\lambda/(\mu + \lambda)) \beta \gamma(i) + (\mu/(\mu + \lambda)) \beta \int_0^{\infty} \max[y, \gamma(i)] \, dF_i(y) \Big\}$$

$$= \max_{0 \leqslant \mu \leqslant \bar{\mu}} \Big\{ -c(\mu) + (\mu/(\mu + \lambda)) \beta \times \Big[\int_0^{\infty} \max[y, \gamma(i)] \, dF_i(y) - \gamma(i) \Big] \Big\} + \beta \gamma(i)$$

$$\equiv \max_{0 \leqslant \mu \leqslant \bar{\mu}} \{ -c(\mu) + (\mu/(\mu + \lambda)) \, \delta(i) \} + \beta \gamma(i),$$

where $\gamma(i) \equiv \sum_{j=1}^{K} P_{ij} V(j)$. Since $V(1) \leqslant V(2) \leqslant \cdots \leqslant V(K)$, it follows that $V(K) \geqslant \gamma(K)$ and $V(1) \leqslant \gamma(1)$. Consequently, we have

$$\begin{aligned} 0 \leqslant V(K) - V(1) &\leqslant \beta\gamma(K) - \beta\gamma(1) + (\mu(K)/(\mu(K) + \lambda))[\delta(K) - \delta(1)] \\ &\leqslant \beta V(K) - \beta V(1) + (\mu(K)/(\mu(K) + \lambda))[\delta(K) - \delta(1)], \end{aligned}$$

or, equivalently,

$$\mu(K)[\delta(K) - \delta(1)] \geqslant (1 - \beta)[\mu(K) + \lambda][V(K) - V(1)].$$

If $\mu(K) = 0$, then $V(K) = V(1)$ and, hence, $\delta(K) = \delta(1)$ so that $\mu(K) = \mu(1)$. On the other hand, if $\mu(K) > 0$, then $\delta(K) - \delta(1) \geqslant 0$, in which case we also have $\mu(K) \geqslant \mu(1)$. If $\beta = 1$, then $\mu(K) = 0$ implies $V(K) - V(1) \leqslant \gamma(K) - \gamma(1)$, from which we can conclude that $V(K) = V(1)$. The case $\mu(K) > 0$ with $\beta = 1$ is exactly the same as when $\beta < 1$. (Note that $V_n(i + 1) - V_n(i)$ nondecreasing in n would suffice here to enable us to assert that $\mu_n(K) \geqslant \mu_n(1)$, all $n \geqslant 1$.) Q.E.D.

In presenting the next two examples, it is convenient to introduce and utilize the continuous-time version of the model which is of some interest in its own right. There is essentially no generalization in allowing the Markov chain which governs the economy to have continuous-time parameter. With the introduction of a variable search intensity, however, the continuous-time formulation permits the searcher, in a very natural and convenient way, to receive a number of offers before there is a change in the state of the economy.

In the continuous-time formulation, we maintain the notion of periods. Periods are defined so that one period begins the moment another period ends, and a period ends when one of the following three events occurs: An offer arrives, the state of the economy changes, a fictitious offer of 0 arrives. At the beginning of each period, the decision maker observes the state (i, x); either he accepts the offer x or he selects a search intensity μ, $0 \leqslant \mu \leqslant \bar{\mu}$, which he maintains during the course of that period.[13]

Associated with search intensity μ is the search cost rate (per unit time) $c(\mu)$ and the exponential time, with parameter μ, till a job offffer arrives. In this formulation, we let the time till the economy changes state and the time till an offer arrives be independent exponential random variables with parameters $\lambda_i \equiv \lambda$ and μ, respectively. For convenience in analysis (see [10] for a complete exposition of this device), it is desirable that the length of each period be a random variable independent of both the state

[13] Since the length of the period turns out to be an exponential random variable, there is no loss of optimality in maintaining a single search intensity as opposed to allowing it to be a function of the time since the period began.

of the system and the decision variables. This is achieved by letting the time till a fictitious offer of 0 arrives be an exponential random variable with parameter $\bar{\mu} - \mu$ that is independent of the two previously defined random variables. Consequently, the length of each period is an exponential random variable with parameter $\Lambda = \lambda + \mu + (\bar{\mu} - \mu) = \lambda + \bar{\mu}$. Thus, Λ is the transition rate of the (two-dimensional—state of economy and offer) Markov chain.

Let (i, x) be the state of the system (Markov chain) at the beginning of the period. If the period ends with (i) an offer, say of y, (ii) a change in the state of the economy, say to j, or (iii) a fictitious offer of 0, then the state at the beginning of the next period will be (i, y), $(j, 0)$, or $(i, 0)$, respectively. As before, $F(\cdot)$ is the offer distribution, and P_{ij} is the probability that the economy will change to j from i given that it changes.

Defining $\alpha > 0$ to be the continuous-time discount factor, the expected discounted cost of search during a period when searching at intensity μ is easily seen to be

$$c(\mu)/(\alpha + \Lambda) = c(\mu) \int_0^\infty \left[\int_0^t e^{-\alpha\tau}\, d\tau\right] \Lambda e^{-\Lambda t}\, dt.$$

Similarly, the expected discounted length of a period is given by

$$\Lambda/(\alpha + \Lambda) = \int_0^\infty e^{-\alpha t} \Lambda e^{-\Lambda t}\, dt.$$

Finally, the probabilities that the period ends with an offer, a change in the state of the economy, or a fictitious offer are μ/Λ, λ/Λ, and $(\bar{\mu} - \mu)/\Lambda$, respectively.

Coupling all of these facts, we obtain the continuous time analog of (6) ($V_0 = 0$)

$$V_{n+1}(i, x) = \max \Big\{ x, (1/(\alpha + \Lambda)) \max_{0 \leqslant \mu \leqslant \bar{\mu}} \Big[-c(\mu) + \lambda \sum_{j=1}^{K} P_{ij} V_n(j, 0) + (\Lambda - \lambda - \mu)\, V_n(i, 0) + \mu \int_0^\infty V_n(i, y)\, dF_i(y) \Big] \Big\}. \tag{10}$$

Before presenting the examples, we note that all of the results in this paper hold for the continuous-time model with the sole exception of Example 1 which requires $n = +\infty$. The proofs are quite similar.

EXAMPLE 3. The maximal search intensity need not be from state K.

Suppose there are three states with F_1 having mass 1 at 0 and F_2 and F_3 having mass 1 at 1. Furthermore, suppose that $c(\mu) = \mu^2$, states 1 and 3 are absorbing, and $P_{21} = 1$. Then for the infinite horizon problem, the return from state 3 when we search at rate μ is

$$(1/\alpha) \cdot (\mu/(\alpha + \mu)) - (1/(\alpha + \mu))\, c(\mu).$$

This yields $\mu(3) = -\alpha + (\alpha^2 + 1)^{1/2}$. Similarly, if search rate μ is employed from state 2, the return

$$(\mu/(\mu + \lambda)) \cdot (1/\alpha) \cdot ((\lambda + \mu)/(\alpha + \lambda + \mu)) - (c(\mu)/(\alpha + \lambda + \mu))$$

is obtained. (The first factor is the probability of eventually getting an offer, the third is the expected discounted time of either receiving an offer or going to state 1, and $1/\alpha$ is the discounted present value of the wage rate 1.) It then follows that

$$\begin{aligned}\mu(2) &= -(\alpha + \lambda) + (\alpha + \lambda)(\alpha^2 + 1)^{1/2}/\alpha \\ &= -\alpha + (\alpha^2 + 1)^{1/2} + \lambda((\alpha^2 + 1)^{1/2}/\alpha - 1) > \mu(3).\end{aligned}$$

Q.E.D.

EXAMPLE 4. The minimal search intensity need not be from state 1.

Suppose that there are three states with F_1 and F_2 having mass 1 at 1 and F_3 having mass 1 at 100. Furthermore, suppose that $c(\mu) = \mu^2$, states 1 and 3 are absorbing, and $P_{21} = P_{23} = \frac{1}{2}$. Then the return from states 1 and 3, respectively, when search intensity μ is employed is

$$V(1) = \frac{1}{\alpha} \cdot \frac{\mu}{\alpha + \mu} - \frac{c(\mu)}{\alpha + \mu} \quad \text{and} \quad V(3) = \frac{100}{\alpha} \cdot \frac{\mu}{\alpha + \mu} - \frac{c(\mu)}{\alpha + \mu},$$

from whence it follows that $\mu(1) = -\alpha + (\alpha^2 + 1)^{1/2}$ and $\mu(3) = -\alpha + (\alpha^2 + 100)^{1/2}$. Similarly, the return from state 2 when rate $\mu(1)$ is used from state 1, $\mu(3)$ from state 3, and μ from state 2 (and, of course, any offer received while in state 2 is accepted) is

$$\begin{aligned}&\frac{\mu}{\lambda + \mu} \cdot \frac{1}{\alpha} \cdot \frac{\lambda + \mu}{\alpha + \lambda + \mu} \\ &\qquad + \frac{\lambda/2}{\lambda + \mu} [V(3) + V(1)] \frac{\lambda + \mu}{\alpha + \lambda + \mu} - \frac{c(\mu)}{\alpha + \lambda + \mu},\end{aligned}$$

so that $\mu(2) = -(\alpha + \lambda) + ((\alpha + \lambda)^2 - \eta)^{1/2}$, where $\eta = (\lambda/2)(V(1) + V(3)) - (\alpha + \lambda)/\alpha$. Finally, it can be seen from these formulas that $\mu(1) > \mu(2)$. Q.E.D.

5. The Discrete Time Model with Resignations

We now extend our model, as embodied in Eq. (3), to include the opportunity of quitting in order to seek a better job. We assume that the wages of employed workers do not change over the business cycle. (See the discussion of this point in Section 1.)

At this point, we must interpret the job offer x as the currently available wage *rate*, so that $x/(1-\beta)$ is the β-discounted present value of working forever (i.e., never resigning) at wage rate x. To begin, define $W_n(i, x)$ to be the maximal β-discounted expected return attainable when n periods remain, the state of the economy is i, and x is the currently available wage rate. Then $W_n(i, x)$ satisfies the recursive equations[14] ($W_0 = 0$)

$$W_{n+1}(i, x) = \max\left\{x + \beta \sum_{j=1}^{K} P_{ij} W_n(j, x), \right.$$

$$\left. -c + \beta \sum_{j=1}^{K} P_{ij} \int_0^\infty W_n(j, y)\, dF_i(y)\right\}. \tag{11}$$

Implicit in (11) is the possibility of not searching via accepting the current wage rate x even if $x = 0$. Of course, the wage rate arising from any past job offer that was either not accepted or accepted but subsequently rejected is no longer available.

As before, we can effect an induction argument based on Assumptions 1 and 2 to verify that $W_n(i, x)$ is nondecreasing in i, x, and n, and, hence, $f_n(i, x)$ and $r_n(i)$ are nondecreasing in i, x, and n, where

$$W_{n+1}(i, x) = \max\{x + f_n(i, x); r_n(i)\}, \tag{12}$$

so that for each i, there is a reservation wage and a resignation wage, and the two are, of course, equal. To verify that the reservation wage is nondecreasing in i and in n, it would suffice to verify that $r_n(i) - f_n(i, x)$ is nondecreasing in i and in n, respectively.

The following example reveals, contrary to intuition, it is not necessarily true that introducing the possibility of resignation lowers the reservation wage. (We conjecture that there is a $J \leqslant K$ such that $W(i) \geqslant V(i)$ if $i \geqslant J$, where $W(i)$ and $V(i)$ are the reservation wages from state i with and without the possibility of resignation, respectively.)

[14] The recursive equation for the continuous time model is ($W_0 = 0$)

$$W_{n+1}(i, x) = (1/(\alpha + \lambda + \bar{\mu})) \max\,(x + \bar{\mu} W_n(i, x) + \lambda \textstyle\sum_{j=1}^{K} P_{ij} W_n(j, x);$$
$$-c + \lambda \textstyle\sum_{j=1}^{K} P_{ij} W_n(j, 0) + \bar{\mu} \int_0^\infty W_n(i, y) dF_i(y)).$$

EXAMPLE 5. The reservation wage can increase with the possibility of resignations.

Take $\alpha = \lambda = \mu = c = 1$ and suppose that F_1 has mass 1 at w_1 and F_2 has mass $\frac{1}{2}$ at w_2 and w_3. We choose $w_1 > 2$ so that it is optimal to search while in state 1. We will choose w_i so that $W(2)$, the reservation wage in state 2, exceeds w_2. To begin, note that using this policy we have[15]

$$\begin{aligned} W(1) &= -\frac{c}{\alpha+\lambda+\mu} + \frac{1}{\alpha+\lambda+\mu} \\ &\quad \times \left\{\mu\left[\frac{w_1}{\alpha+\lambda} + \frac{\lambda}{\alpha+\lambda} W(2)\right] + \lambda W(2)\right\} \\ &= -\frac{c}{\alpha+\lambda+\mu} + \frac{\mu}{\alpha+\lambda+\mu}\frac{w_1}{\alpha+\lambda} + \frac{\lambda}{\alpha+\lambda} W(2), \end{aligned}$$

and

$$W(2) = -\frac{c}{\alpha+\lambda+\mu} + \frac{1}{\alpha+\lambda+\mu}\left\{\mu\left[\frac{W(2)}{2} + \frac{w_3}{2\alpha}\right] + \lambda W(1)\right\},$$

so that

$$W(2) = -(2/3) + (w_3/4) + (w_1/12).$$

Setting $w_1 = 8$, $w_2 < 12$, and $w_3 = 48$, we obtain $W(2) = 12 > w_2$, so the reservation wage from state 2 is, indeed, w_3.

When resignation is not permitted, we need consider two policies: accept w_1 and w_3 or accept only w_3.

If we accept w_1 and w_3, we are lead to

$$V(1) = -\frac{c}{\alpha+\lambda+\mu} + \frac{w_1}{\alpha}\frac{\mu}{\alpha+\lambda+\mu} + V(2)\frac{\lambda}{\alpha+\lambda+\mu},$$

and

$$V(2) = -\frac{c}{\alpha+\lambda+\mu} + \frac{1}{\alpha+\lambda+\mu}\left\{\mu\left[\frac{1}{2}V(2) + \frac{1}{2}\frac{w_3}{\alpha}\right] + \lambda V(1)\right\},$$

so that $V(2) = 152/13 < 12$.

If we accept only w_3, then the equation for $V(1)$ becomes $V(1) = -c/(\alpha+\lambda) + \lambda V(2)/(\alpha+\lambda)$ if search is required and $V(1) = \lambda V(2)/(\alpha+\lambda)$ if search is not required. This leads to $V(2) = 45/4$ and $V(2) = 46/4$, respectively.

Consequently, setting $w_2 > 152/13$ (but also $w_2 < 12$), we see that w_2 exceeds $V(2)$, the reservation wage in state 2. Q.E.D.

[15] Here, $W(i)$ is the infinite horizon return starting from state i when an offer is not yet in hand. Similarly for $V(i)$.

At first blush, it would seem evident that the reservation wage is monotone increasing even when the resignation opportunity is included. The following example, however, illustrates the refractory nature of this particular embellishment of the basic model.

EXAMPLE 6.[16] The reservation wages need not be nondecreasing in the state.

Suppose there are three states with $F_1 = F_2 = \frac{1}{2}(\delta_1 + \delta_4)$ and $F_3 = \delta_{100}$, where δ_x is the point mass at x. Let states 1 and 3 absorbing and $P_{22} = p = 1 - P_{23}$ with $p = 1/10$ and take $c < 5/28$ and the discrete time discount factor $\beta = \frac{2}{3}$. We consider the discrete time infinite horizon problem. The sequencing of events within a period is as follows: search, job offer, transition according to $\{P_{ij}\}$, and then collect the wage at the beginning of the next period.

From state 1 there are but two policies to follow: accept both 1 and 4 or accept only 4. The return for these policies satisfy, respectively, the equations

$$W = -c + \beta(\tfrac{1}{2}1 + \tfrac{1}{2}4)/(1 - \beta) = 5 - c$$

and

$$W = -c + \beta\left(\frac{1}{2}\frac{4}{1-\beta} + \frac{1}{2}W\right), \quad \text{i.e.,} \quad W = 5 - c + \frac{1}{2}(2 - c).$$

Thus, the reservation wage from state 1 is 4.

To find the reservation wages associated with the above policies for state 2, we begin by letting X (roughly the discounted value of wages earned while residing in state 2 if an offer of 1 is accepted) satisfy

$$P(X = 1 + \beta + \cdots + \beta^n) = (1 - p)\,p^n.$$

Then $E(X) = 1 + p\beta E(X)$ so that $E(X) = 1/(1 - p\beta)$. The discounted value of the net benefits while in state 2 under the two policies satisfy, respectively,

$$W = -c + \beta\{p(5/2)\,E(X) + (1 - p)\,0\},$$

and

$$\begin{aligned} W &= -c + \beta\{p[\tfrac{1}{2} \cdot 4E(X) + \tfrac{1}{2}W] + (1 - p)\,0\} \\ &= -c + 2E(X)\,\beta p + \beta p W/2, \end{aligned}$$

i.e.,

$$W = (2/(2 - \beta p))\{2\beta p E(X) - c\}.$$

As $E(X) = 15/14$, this yields values of $(5/28) - c$ and $(30/29)((4/28) - c)$, so the reservation wage from state 2 is 1. Q.E.D.

[16] Roughly, the counterexample shows that if $F_1 = F_2 = \cdots = F_{K-1} \ll F_K$ and $P_{ij} = 0$ for $j < i$, then we can have the reservation wages decrease ($R(1) \geqslant R(2) \geqslant \cdots \geqslant R(K - 1)$) until state K is reached (with $R(K) \gg R(1)$).

Once again, the special case $P_{ij} = P_j$ is tractable. Here, we note that

$$f_n(i, x) = \beta \sum_{j=1}^{K} P_{ij} W_n(j, x) = \beta \sum_{j=1}^{K} P_j W_n(j, x)$$

is independent of i, so that $r_n(i) - f_n(i, x)$ is, indeed, nondecreasing in i. Thus, the conclusion of nondecreasing reservation wages does hold for this special case.

Retaining the assumption that $P_{ij} = P_j$, we can show that the quit rate varies directly with the state of the economy. Consequently, as the state of the economy increases three things happen: Quits rise, the period of frictional unemployment (stochastically) decreases, and labor force participation increases. The net effect of these forces on the unemployment rate is unclear. Whatever the relationship is, it is not the Phillips curve in the usual sense. Here we are postulating real improvements in the economy; that is, we are concerned with the relationship between the change in the real rate of growth and unemployment. The state of the economy is known to everyone, thereby obviating any expectations model.

It has been observed [3] that the probability of resignation decreases the longer a worker has been employed by a given firm (or at least a given industry). The explanation posited rests upon the notion that while employed a worker accumulates specific human capital. The worker's wage reflects this investment; that is, it includes a return on the worker's investment in specific human capital. The important point is that this investment is not transferable. Thus the worker incurs a capital loss when he quits his current job.[17]

To incorporate the wage paid on human capital, let $\epsilon(k) \geqslant 0$ be the incremental wage paid on human capital to a worker who has been on the job for k periods. Define $W_n(i, x, k)$ to be the maximal β-discounted expected return attainable when n periods remain, i is the state of the economy, x is the current available wage rate, and k is the number of periods the searcher has been employed at his current job ($k = 0$ means that he is currently unemployed). Clearly, $W_n(i, x, k)$ satisfies ($W_0 = 0$)

$$\begin{aligned} W_{n+1}(i, x, k) &= \max \left\{ x + \beta \sum_{j=1}^{K} P_{ij} W_n(j, x + \epsilon(k), k + 1); \right. \\ &\qquad \left. -c + \beta \sum_{j=1}^{K} P_{ij} \int W_n(j, y, 0)\, dF_i(y) \right\} \\ &= \max\{x + w_n(i, x, k); r_n(i)\}. \end{aligned} \tag{13}$$

[17] We assume here that the worker was the sole investor in this firm specific capital. In practice, firms and workers tend to share in this investment. Consequently, firms are less inclined to fire. See [18].

Defining $R_{n+1}(i, k)$ to be the unique value of x for which $x = r_n(i) - w_n(i, x, k)$, we can conclude that $R_n(i, k)$ is the *resignation wage* when n periods remain, i is the state of the economy, and k is the number of periods the searcher has been employed. (Thus, $R_n(i, 0)$ is the reservation wage.) To see that $R_{n+1}(i, k)$ exists and is uniquely defined, it suffices (see Fig. 1) to show that $w_n(i, x, k)$ is a nondecreasing function of x. Once again, a straightforward induction argument utilizing Assumptions 1 and 2 verifies that $W_n(i, n, k)$, and, hence, $w_n(i, x, k)$, are nondecreasing in i and x. Moreover, if $\epsilon(k)$ is a nondecreasing [nonincreasing] function of k, then so is $W_n(j, x, k)$. Consequently, the resignation wage $R_n(i, k)$ is a nonincreasing [nondecreasing] function of k if $\epsilon(k)$ is a nondecreasing [nonincreasing] function of k. (Note too that if $P_{ij} = P_j$, $R_n(i, k)$ is nondecreasing in i since $w_n(i, x, k)$ is independent of i.)

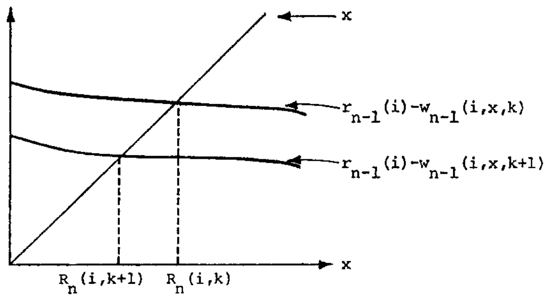

FIGURE 1

Presumably, $\epsilon(k)$ increases with k, or equivalently, specific human capital increases, so our model predicts the anticipated result that the resignation wage decreases with time on the job. That is, the longer the time on the job, the more reluctant the worker is to resign (equivelently, the smaller the wage needed to induce him to remain on the job).

6. THE DISCRETE TIME MODEL WITH FIRING

We now admit the possibility of being fired. This is incorporated by assuming that the probability of being fired is f_i, where i is the state of the economy. Note that f_i does not depend on the current wage. Letting $(i, 0)$ and $(i, 0^-)$ be the state of the system when the economy is in state i and the worker is not employed or employed at wage rate 0, respectively, we obtain the recursive equations[18] ($U_0 \equiv 0$)

$$U_{n+1}(i, x) = \max \left\{ x g_{n+1}(i) + \beta \sum_{j=1}^{K} P_{ij}[f_i U_n(j, 0) + (1 - f_i) U_n(j)]; \right.$$

$$\left. - c + \beta \sum_{j=1}^{K} P_{ij} \int_0^\infty U_n(j, y)\, dF_i(y) \right\}, \tag{14}$$

where

$$U_{n+1}(i) = \beta \sum_{j=1}^{K} P_{ij}[f_i U_n(j, 0) + (1 - f_i) U_n(j)], \tag{15}$$

and

$$U_{n+1}(i, 0^-) = \beta \sum_{j=1}^{K} P_{ij}[f_i U_n(j, 0) + (1 - f_i) U_n(j, 0^-)], \tag{16}$$

and $g_{n+1}(i)$ is the expected discounted time employed when $n + 1$ periods remain and the initial state is i. As given in (14), (15), and (16), our formulation permits the searcher to work at the wage 0 in order to avoid the search cost; however, search cannot be resumed until he is fired (from this job with wage rate 0). Clearly, there is a reservation wage for each state of the economy.

7. The Discrete Time Model with Resignations and Firing

We now combine the two possibilities of quits and layoffs. As before, let f_i be the firing probability associated with state i and let $Z_n(i, x)$ denote the maximal β-discounted expected return attainable when n periods remain,

[18] The recursive equation for the continuous time model is ($U_0 \equiv 0$)

$$\begin{aligned} U_{n+1}(i, x) = (1/(\alpha + \Lambda)) \max (&x \cdot g_{n+1}(i) + \lambda \sum P_{ij} U_n(j) \\ &+ f_i U_n(i, 0) + (\Lambda + \lambda - f_i) U_n(i); \\ &- c + \lambda \sum P_{ij} U_n(j, 0) \\ &+ (\Lambda - \lambda - \mu) U_n(i, 0) + \mu \textstyle\int U_n(i, y) dF_i(y)), \end{aligned}$$

where

$$U_{n+1}(i) = (1/(\alpha + \Lambda))(\lambda \sum P_{ij} U_n(j) + f_i U_n(i, 0) + (\Lambda - \lambda - f_i) U_n(i)),$$

$\Lambda = \lambda + \mu + \max(f_i)$, and f_i, λ, μ are the parameters of the exponential time till being fired, change of state, and arrival of an offer, respectively, and $g_n(i)/(\alpha + \Lambda)$ is the expected discounted time employed when n periods remain and the initial state is i.

the state of the economy is i, and x is the current available wage rate. Then $Z_n(i, x)$ satisfies[19] ($Z_0 \equiv 0$)

$$\begin{aligned} Z_{n+1}(i, x) &= \max \Big\{ x + \beta \sum_{j=1}^{K} P_{ij}[f_i Z_n(j, 0) + (1 - f_i)\, Z_n(j, x)]; \\ &\qquad -c + \beta \sum_{j=1}^{K} P_{ij} \int_0^{\infty} Z_n(j, y)\, dF_i(y) \Big\} \\ &\equiv \max\{z_n(i, x); \rho_n(i)\}. \end{aligned} \tag{17}$$

As before, the existence of a reservation wage (that also equals the resignation wage) can easily be established inductively by showing that $Z_n(j, x)$ is nondecreasing in x.

8. Conclusion

This paper has constructed a model of job search in a changing economy. As expected, the reservation wage increased as the economy moved to improved states. However, only under fairly stringent assumptions on the stochastic process were we able to say anything definite about the duration of search unemployment in relation to the changing economy. Similarly, it was shown that the optimal search intensity need not increase with the state of the economy unless the same restrictions were imposed. Quits and layoffs were also considered in this dynamic setting, and the existence of reservation and resignation wages was established; however, the further assumption that $P_{ij} = P_j$ was required to ensure that the reservation wage is monotone increasing in the state of the economy.

There are several directions in which these models can be modified. First, the treatment of specific human capital can be extended to the model with both resignations and layoffs with the relative contributions of employer and employee to the investment being explicitly considered. This should yield testable hypotheses with respect to quit and layoff behavior over the business cycle. Second, we have emphasized the difficulties associated with our models of quits and layoffs. Presumably, firms will be adjusting the wages of employees as the economy changes. We did not attempt to model this adjustment process, but merely assumed that

[19] The recursive equation for the continuous time model is ($Z_0 \equiv 0$, $\Lambda = \lambda + \mu + \max(f_i)$)

$$\begin{aligned} Z_{n+1}(i, x) = (1/(\alpha + \Lambda)) \max(&x + \lambda \sum P_{ij} Z_n(j, x) + f_i Z_n(i, 0) + (\Lambda - \lambda - f_i) Z_n(i, x); \\ &-c + \lambda \sum P_{ij} Z_n(j, 0) + (\Lambda - \lambda - \mu) Z_n(i, 0) + \mu \textstyle\int Z_n(i, y) dF_i(y)). \end{aligned}$$

wages were constant. Clearly, additional work is required here. Third, the interaction of optimal employer and employee policies gives rise to equilibrium wage distributions. (See [4, 12, 16, 27] for analyses of equilibrium wage distributions in a static setting.) Analysis of an endogenous wage distribution in a dynamic setting strikes us as an extremely important topic for additional research. It is precisely models like these which will provide a rigorous foundation for the study of macroeconomic phenomena. Finally, we have assumed throughout that job searchers know the stochastic structure of the process. In practice, they will know neither the wage distributions nor the Markov transition probabilities. The reformulation of these models to include adaptation is another area for future research.

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